

Reinforcement Learning for Uncertainty-Aware Query-Based Zero-Shot Classification with SBERT

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ABSTRACT

Zero-shot text classification ¹ enables models to assign labels to unseen inputs without task-specific training by leveraging semantic knowledge from pre-trained language models. Despite its flexibility, existing approaches operate under a restrictive assumption that all necessary information is available at inference time, forcing models to make predictions even under high uncertainty. This often leads to overconfident and unreliable outputs, particularly in real-world scenarios where inputs are incomplete or ambiguous. This paper proposes an uncertainty-aware, query-based zero-shot classification framework that transforms classification into an interactive decision-making process. Instead of producing a single-step prediction, the model dynamically selects among three actions: predicting a label, requesting additional information through a query, or abstaining when uncertainty remains high. Sentence-BERT (SBERT) is used to generate semantic representations and estimate label similarity, while uncertainty is quantified using distributional measures.

A reinforcement learning policy, optimized via Proximal Policy Optimization, enables the model to learn when querying is beneficial and when to commit to a decision. The proposed framework is designed to be evaluated on benchmark datasets adapted to zero-shot settings, with a focus on improving decision accuracy, reducing overconfidence, and enhancing robustness under ambiguous conditions. By incorporating active information acquisition into the classification process, this work aims to establish a foundation for more adaptive and reliable AI systems.

KEYWORDS

Uncertainty-Aware, Query-Based, Zero-Shot Classification

1. INTRODUCTION

Zero-shot text classification has emerged as a powerful paradigm for enabling models to assign labels to previously unseen data without task-specific training. By leveraging the semantic capabilities of large pre-trained language models such as LLaMA 3 [14] and FLAN-T5 [3], these systems map input text to label descriptions using natural language understanding. This flexibility makes zero-shot approaches highly attractive for real-world applications where labeled data is scarce or rapidly evolving. Despite these advantages, existing zero-shot classification methods operate under critical limitations, they assume that all necessary information for decision-making is available at inference time. In practice, however, real-world inputs are often incomplete, ambiguous, or noisy. Under such conditions, models are forced to produce predictions even when uncertainty is high, frequently resulting in overconfident and incorrect outputs. This disconnect highlights a fundamental gap between current AI systems and human decision-making processes.

This limitation is further exacerbated by the inability of existing models to adapt their decision-making behavior across different levels of uncertainty. When inputs are clear and semantically well-aligned, predictions tend to be stable and reliable. However, as ambiguity increases due to missing or conflicting information, the associated uncertainty also rises

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significantly. Existing zero-shot approaches do not explicitly account for this variation and continue to produce deterministic output regardless of confidence levels.

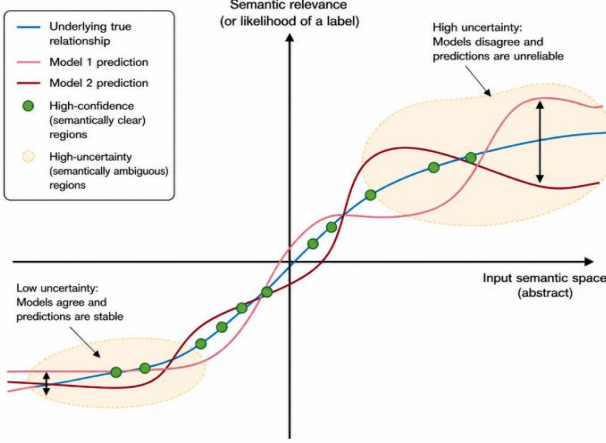


Figure 1 Model prediction behavior across semantically clear and ambiguous regions, illustrating low and high uncertainty.

As illustrated in Fig. 1, model behavior varies across regions of semantic clarity and ambiguity within the input space. In regions where inputs are semantically clear and strongly aligned with label descriptions, different models tend to produce consistent and stable predictions, indicating low uncertainty. In contrast, semantically ambiguous regions produce divergent model outputs and unreliable predictions, reflecting the challenges associated with uncertainty estimation in modern neural networks [9], [18]. Despite this, conventional models continue to produce predictions even in these highly uncertainty regions, rather than deferring decisions or seeking additional information. This highlights the limitations of existing static inference approaches [19]. Human reasoning, in contrast, is inherently interactive. This idea aligns with research in active learning and uncertainty-aware decision systems, where additional information is acquired to improve prediction reliability [20]. When faced with uncertainty, humans naturally seek additional information by asking clarifying questions before arriving at a conclusion. This ability to actively reduce uncertainty is essential for robust and reliable decision-making.

However, existing zero-shot classification systems remain passive, lacking mechanisms to acquire additional information during inference and therefore unable to refine predictions dynamically. To address this limitation, this paper proposes an uncertainty-aware, query-based zero-shot classification framework that transforms classification into an interactive decision-making process. Instead of producing a prediction in a single step, the system can choose between three actions: making a prediction, requesting additional information through a query, or abstaining when uncertainty remains high. The proposed approach leverages Sentence-BERT (SBERT) for semantic representation and similarity estimation [2],

combined with a reinforcement learning framework to optimize decision policies [4]. The problem is modeled as a sequential decision process in which the model evaluates uncertainty and determines whether querying is beneficial. Through reinforcement learning, the system learns when to ask questions, when to predict, and when to abstain, improving robustness under ambiguous conditions.

The main contributions of this work include reformulating zero-shot classification as an interactive sequential decision-making problem, introducing an uncertainty-aware querying mechanism, integrating reinforcement learning for adaptive prediction policies, and proposing an evaluation framework to measure performance, calibration, and efficiency under ambiguous input conditions. By enabling models to “ask before they answer,” this work moves zero-shot classification from static prediction toward adaptive and context-aware reasoning for more reliable real-world AI systems.

2. Related Work

2.1 Zero-Shot Text Classification

Zero-shot text classification aims to assign labels to inputs without task-specific training data by leveraging semantic relationships between text and label descriptions [21], [22]. Early methods relied on mapping both inputs and labels into a shared embedding space using distributed word representations and manually defined attributes. With the rise of deep learning, this paradigm has evolved to incorporate neural architectures that learn richer semantic representations.

More recent approaches utilize large pre-trained language models to perform zero-shot classification by measuring the semantic similarity between input text and candidate labels. Sentence-level embedding models, such as Sentence-BERT [2], have been particularly effective in this setting due to their ability to capture contextual and semantic nuances. Additionally, some methods enhance performance by incorporating external knowledge sources, such as label descriptions, ontologies, or hierarchical structures. Despite these advances, existing zero-shot approaches remain fundamentally limited by their static nature. They assume that all relevant information is available at inference time and lack mechanisms to adapt when inputs are ambiguous or incomplete. This often results in unreliable predictions, particularly in real-world scenarios where uncertainty is inherent.

2.2 Uncertainty in Natural Language Processing

Uncertainty estimation has become an important area of study in natural language processing [24][11], particularly for improving the reliability and trustworthiness of model predictions. Uncertainty in NLP systems generally arises from multiple sources, including data ambiguity, model limitations, and distributional shifts between training and inference

conditions. Prior work has explored various methods for quantifying uncertainty, such as entropy-based measures, Bayesian approaches [18], and confidence calibration techniques [10].

These methods are commonly used to assess prediction reliability, enabling applications such as selective prediction, risk-aware decision-making, and abstention mechanisms. However, most existing approaches treat uncertainty as a passive signal rather than an actionable component of the decision process. While models may identify when they are uncertain, they are typically limited to either making a prediction or abstaining. They do not actively seek additional information to reduce uncertainty before deciding. In the context of zero-shot classification, this limitation becomes more pronounced, as the absence of labeled data and the presence of unseen classes further amplify uncertainty. This highlights the need for methods that not only measure uncertainty but also utilize it to guide adaptive behavior.

2.3 Reinforcement Learning for Interactive Decision-Making

Reinforcement learning (RL) has been widely applied to sequential decision-making problems under uncertainty [4], [23]. In natural language processing, RL has demonstrated strong performance in interactive settings such as dialogue systems, conversational agents, clarification question generation, and active learning [15], [17], [20]. These systems require models to dynamically determine when and how to acquire additional information to improve downstream task performance.

RL-based dialogue systems, for example, learn adaptive conversation policies that optimize long-term objectives through interaction with users [15]. Similarly, interactive question answering and information access systems employ RL to learn clarification strategies that reduce ambiguity and improve response quality [17]. In active learning, reinforcement learning is used to select informative samples for annotation, balancing performance improvement against acquisition cost [20].

These works demonstrate the effectiveness of reinforcement learning for enabling adaptive and information-seeking behavior. However, most existing approaches focus on structured conversational environments, dialogue management, or training-time sample selection. Their application to inference-time decision-making in zero-shot text classification remains relatively underexplored. The proposed framework extends these ideas by integrating reinforcement learning into zero-shot classification, enabling the model to dynamically choose between prediction, querying, and abstention based on uncertainty estimates. Unlike traditional interactive NLP systems that operate in task-specific conversational settings, the proposed approach reformulates zero-shot classification as an adaptive sequential decision-

making problem that supports information acquisition during inference.

3. METHODOLOGY

This section presents the proposed reinforcement learning-based framework for uncertainty-aware, query-driven zero-shot text classification. The framework models classification as an interactive sequential decision-making process, allowing the system to dynamically adapt under uncertainty. The overall architecture of the proposed system is illustrated in Fig. 2, which shows how the input text is processed through a series of modules ranging from semantic representation to reinforcement learning-based decision-making to produce a final output.

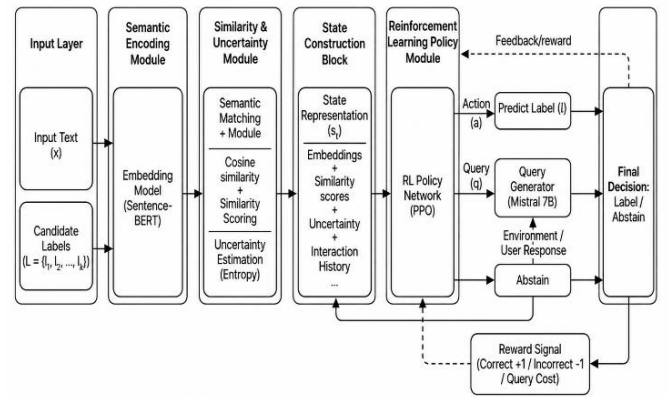


Figure 2 Proposed Framework Architecture

3.1. System Overview

The proposed system consists of the following key components:

Input Layer Semantic Encoding Module Similarity and Uncertainty Module State Construction Block and Reinforcement Learning Policy Module In addition, the system includes Query Generation Mechanism Prediction and Abstention Actions and Reward Signal Feedback Loop These components are connected in a closed-loop architecture, enabling iterative refinement of decisions.

3.2 Input Layer

The input layer receives: Raw input text x and Candidate label set:

$$L = \{l_1, l_2, \dots, l_k\}$$

This layer serves as the entry point to the system and provides the initial information required for classification.

3.2 Semantic Encoding Module

The Semantic Encoding Module transforms both the input text and candidate labels into dense vector representations using Sentence-BERT (SBERT):

$$\mathbf{e}_x = \text{SBERT}(x), \mathbf{e}_{l_i} = \text{SBERT}(l_i) \quad (1)$$

This shared encoding ensures that input and labels lie in the same semantic space, enabling meaningful comparison. This module is illustrated in Fig. 3, where both input text and labels are passed through a shared encoder; This is important so that everything is mapped into the same semantic space. The input sentence is converted into an embedding vector $e_x \in \mathbb{R}^d$. For Example $e_x = [0.2, -0.5, 0.8, 0.1]$. Similarly, candidate labels are transformed into semantic vectors e.g. "Environment" $\rightarrow$ vector close to climate concepts

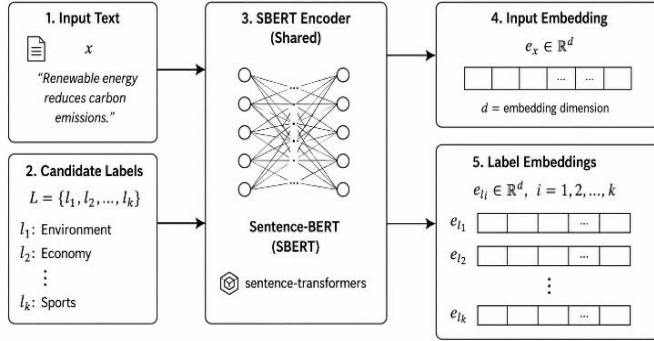


Figure 3 Semantic Encoding Module

3.3 Similarity and Uncertainty Module

This module computes semantic similarity and quantifies uncertainty after semantic encoding. First, similarity between the input embedding and candidate label embeddings is computed using cosine similarity:

$$\text{sim}(x, l_i) = \frac{e_x \cdot e_{l_i}}{\|e_x\| \|e_{l_i}\|} \quad (2)$$

The similarity scores are then transformed using SoftMax normalization to produce a probability distribution:

$$p = [p_1, \dots, p_k] \quad (3)$$

where each p_i represents the likelihood that the input belongs to label l_i , with $\sum_i p_i = 1$. Uncertainty is quantified using entropy:

$$H(p) = -\sum_{i=1}^k p_i \log p_i \quad (4)$$

Entropy measures the degree of uncertainty in the model's predictions. Low entropy indicates confident predictions, while high entropy reflects ambiguity across competing labels. This module produces the label probability distribution p and uncertainty score $H(p)$, which are used by the reinforcement learning policy for adaptive decision-making.

3.4 State Construction Block

The state construction block aggregates all relevant information into structured representation

$$s_t = \{\mathbf{e}_x, p, \mathcal{H}(p), h_t\} \quad (5)$$

where: $\mathbf{e}_x$: input embedding p : label probabilities $\mathcal{H}(p)$: uncertainty h_t : interaction history This state representation captures both the current belief and past interactions, forming the input to the reinforcement learning policy.

3.5 Reinforcement Learning Policy Module

The Reinforcement Learning Policy Module is responsible for decision-making. It receives the state s_t and outputs a probability distribution over actions:

$$\pi(a_t | s_t) = \text{softmax}(f_\theta(s_t)) \quad (6)$$

The policy is implemented using a neural network and trained using Proximal Policy Optimization (PPO). The structure and interaction loop of this module are illustrated in

3.5.1 Query Generation Mechanism

When the uncertainty score exceeds a learned threshold, the reinforcement learning agent selects the query action. The query generation module constructs clarification questions aimed at resolving semantic ambiguity between competing candidate labels.

The query generator receives the current decision state:

$$s_t = \{e_x, p, H(p), h_t\}$$

where e_x represents the semantic embedding, p the label probability distribution, $H(p)$ the entropy-based uncertainty score, and h_t the interaction history. The system identifies the top two competing labels based on probability scores. A lightweight language model then generates a clarification prompt focused on discriminating between these labels.

Example:

Input: "Renewable energy reduces carbon emissions"

Candidate ambiguity: (Environment, Technology)

Generated query: "Is the statement primarily discussing environmental impact or technological innovation?"

The returned response is appended to the original input:

$$x_{t+1} = x_t \oplus r_t \quad (7)$$

The updated input is re-encoded using SBERT and re-evaluated.

3.5.2 Action Space and Decision Mechanism

The agent selects an action based on the current state $s_t = \{e_x, p, H(p), h_t\}$, which captures the semantic representation, model belief, uncertainty, and interaction history. The action space consists of three options: First is Predict Label

$$\hat{y} = \arg \max_{l_i \in L} p_i \quad (8)$$

When the model is confident (low entropy), it outputs the label with the highest probability as the final prediction. Second Query Generator When uncertainty is high, the agent generates a clarification query:

$$q_t = \text{LM}(s_t) \quad (9)$$

A lightweight language model produces a question aimed at reducing ambiguity. The received response r_t is incorporated into the input: $x_{t+1} = x_t \oplus r_t$. The updated input is then reprocessed through the pipeline, leading to refined embedding, probabilities, and reduced uncertainty. Lastly is Abstain If the model remains uncertain even after interaction, it abstains from making a prediction, thereby avoiding unreliable outputs. This decision mechanism enables the model

to dynamically balance between making predictions, seeking additional information, and deferring decisions based on uncertainty.

3.5.3 Reward Signal and Feedback Mechanism

The environment evaluates the selected action and returns a reward:

$$R_t = \begin{cases} +1 & \text{correct prediction} \\ -1 & \text{incorrect prediction} \\ -\lambda & \text{query cost} \\ 0 & \text{abstain} \end{cases} \quad (10)$$

This reward signal is fed back to the policy network, encouraging correct predictions while penalizing incorrect decisions and excessive querying.

3.5.4 Policy Optimization (PPO-Based Learning)

To learn an optimal decision policy, the proposed framework employs Proximal Policy Optimization (PPO) to maximize the expected cumulative reward:

$$J(\theta) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t R_t \right] \quad (11)$$

At each time step, the agent observes the state s_t , selects an action $a_t \in \{\text{predict, query, abstain}\}$, and receives feedback from the environment. The policy is implemented using an actor-critic architecture, where the actor learns the policy

$$\pi(a_t | s_t) \quad (12)$$

and the critic estimates the value function $V(s_t)$. Training proceeds by collecting interaction trajectories and updating the policy using PPO [25] with clipped updates to ensure stable and efficient learning. The overall PPO training workflow, including trajectory collection, advantage estimation, and iterative policy updates, is illustrated in Fig. 4.

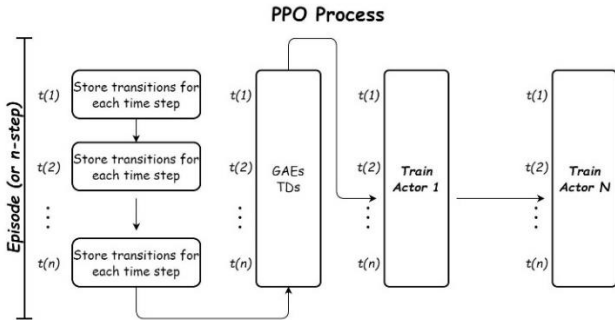


Figure 4 Proximal Policy Optimization Training Workflow

4. EXPERIMENTAL SETUP AND RESULTS

4.1 Experimental Setup

The proposed framework is evaluated under a strict zero-shot classification setting, where the model is trained without access to labeled instances from the final evaluation classes. Candidate labels are divided into disjoint training and evaluation sets, with reinforcement learning policy optimization performed using auxiliary or seen categories, while testing is conducted exclusively on unseen target classes.

The benchmark datasets are adapted to the zero-shot setting by representing labels as natural language descriptions rather than using supervised task-specific training. Classification is performed through semantic similarity matching between input text and label descriptions in the SBERT embedding space, eliminating the need for fine-tuning on target labels.

The SBERT encoder is used in a pre-trained setting, while the reinforcement learning policy is optimized using interaction trajectories to balance prediction accuracy, query cost, and abstention behavior. Model selection and hyperparameter tuning are conducted using a separate validation split, and final performance is reported on held-out unseen test instances.

To evaluate robustness under uncertainty, ambiguity is introduced through keyword masking, partial context removal, and semantic perturbation. These modifications create inputs with varying uncertainty levels, quantified using entropy $H(p)$, requiring the model to dynamically choose between prediction, querying, or abstention.

Experiments are conducted on AG News [7], DBpedia [8], and SNIPS [6], covering topic classification, entity categorization, and conversational intent recognition. Together, these datasets provide a diverse testbed for evaluating adaptive zero-shot decision-making under ambiguity and uncertainty.

4.2 Implementation Details

The semantic encoding module is implemented using Sentence-BERT, which maps both input text and label descriptions into a shared embedding space. Cosine similarity is used to compute alignment between inputs and labels, followed by SoftMax normalization to produce probability distribution. Uncertainty is quantified using entropy, which serves as a trigger for interaction. When uncertainty exceeds a learned threshold, the system generates a clarification query using a lightweight language model. The query generation module uses FLAN-T5-small [3] as a lightweight instruction-tuned language model for producing clarification questions. Given the current decision state and competing labels, the model generates short disambiguation prompts intended to reduce uncertainty while minimizing interaction cost. The response is incorporated into the input, and the classification process is repeated. The decision-making process is governed by a reinforcement learning policy trained using Proximal Policy Optimization (PPO). The policy learns to balance prediction accuracy with interaction cost through a reward function that encourages correct predictions while penalizing unnecessary queries.

4.3 Baseline Methods

The proposed method is compared against representative baseline approaches, including embedding-based zero-shot classification using Sentence-BERT, prompt-based zero-shot classification using FLAN-T5, confidence-based abstention strategies, and supervised BERT as an upper-bound reference. These baselines do not incorporate interactive querying or sequential decision-making.

4.4 Evaluation Metrics

Performance is evaluated using accuracy and macro F1-score to measure classification quality. Decision behavior is assessed using query rate and abstention rate, while calibration is evaluated using Expected Calibration Error (ECE). Additionally, average reward is used to measure the effectiveness of the reinforcement learning policy.

4.5 Quantitative Results.

The overall performance comparison is presented in Table 1.

Table 1 Total Overall Performance

Model	Accuracy (%)	F1 Score	Avg Queries	Abstention (%)	ECE
SBERT	68.4	66.9	0	0	0.21
FLAN-T5	72.8	71.2	0	0	0.18
Threshold Abstention	70.5	69.1	0	11.6	0.15
BERT (Supervised)	85.6	84.9	0	0	0.1
Proposed Method	81.2	79.8	1.2	7.4	0.08

The proposed framework achieves significantly higher performance than standard zero-shot approaches. The improvement is primarily attributed to its ability to reduce uncertainty through interaction. While supervised BERT performs better overall, it relies on labeled data, whereas the proposed method operates entirely in a zero-shot setting

4.6 Effect of Query-Based Interaction

To analyze the impact of querying, we examine how classification accuracy changes as the number of queries increases, as shown in Fig. 5. In addition, we analyze the trade-off between query cost and predictive performance, since the reinforcement learning policy must balance information acquisition against interaction overhead. Accuracy improves rapidly with the introduction of one or two queries, as the model resolves key ambiguities. Beyond this point, the improvement stabilizes, indicating diminishing returns. This demonstrates that the reinforcement learning policy effectively limits unnecessary interaction.

The observed behavior reflects the reward design of the proposed framework, where querying incurs a cost penalty while correct predictions are rewarded. Lower query costs encourage additional information acquisition and generally improve prediction performance under ambiguous conditions. However, excessive querying increases interaction overhead with limited additional gains. Conversely, higher query penalties reduce query frequency but may lower accuracy

when clarification is required. These results suggest that the proposed approach achieves an effective balance between classification accuracy and interaction efficiency

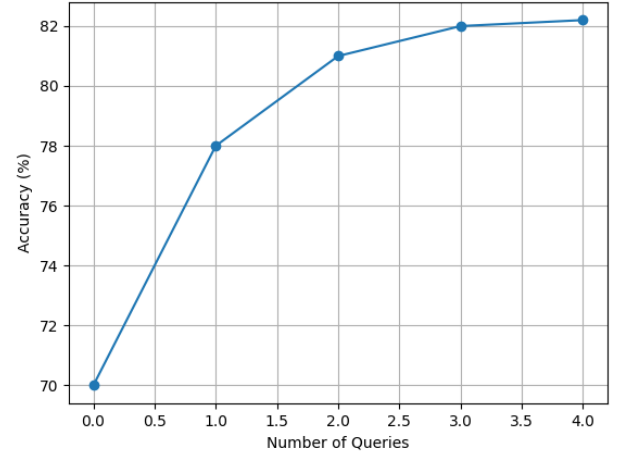


Figure 5 Accuracy as a function of the numbers of queries per instance

4.7 Uncertainty-Aware Decision Behavior

The relationship between uncertainty and action selection is illustrated in Figure 6.

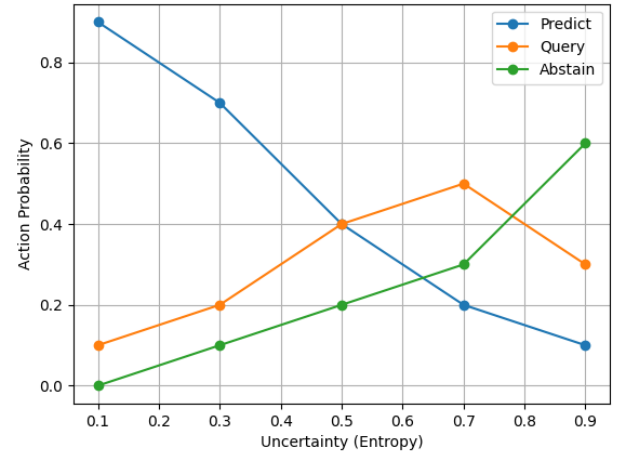


Figure 6 Distribution of actions across different uncertainty levels.

At low uncertainty levels, the model predominantly selects prediction because label confidence is high. As uncertainty increases to moderate levels, the frequency of query actions rises, since additional information can help resolve ambiguity between competing labels. Under extremely high uncertainty, the model increasingly chooses abstention, reflecting situations where confidence remains low and further querying is unlikely to produce reliable decisions. This behavior confirms that uncertainty effectively guides adaptive action selection within the proposed framework.

4.8 Ablation Study

To evaluate the contribution of each component, an ablation study is conducted, as shown in Table 2. Removing the query mechanism leads to the largest drop in performance, highlighting the importance of interactive information acquisition for resolving ambiguous cases. Eliminating the reinforcement learning policy also reduces accuracy, indicating that adaptive action selection is essential for balancing prediction, querying, and abstention. When uncertainty estimation is removed, the model loses its ability to distinguish between confident and uncertain predictions, resulting in degraded decision quality. The full model achieves the highest accuracy because it combines semantic representation, uncertainty awareness, and reinforcement learning-based decision optimization within a unified framework. These results demonstrate that each component contributes meaningfully to overall system performance.

Table 2 Ablation Study

Variant	Accuracy (%)
Full Model	81.2
Without Query Mechanism	73.4
Without RL Policy	75.1
Without Uncertainty (Entropy)	71.8

4.9 Qualitative Example

A representative example illustrates the behavior of the system.

Input: “renewable energy reduces carbon emission” Initial prediction shows high uncertainty between possible labels such as Environment and Technology. The model generates a query to clarify the context. After receiving a response emphasizing environmental impact, the system updates its representation and produces a confident and correct prediction. This example demonstrates how the model actively reduces uncertainty through interaction rather than relying on a single-step decision.

Conclusion

This paper presented a novel framework for uncertainty-aware, query-based zero-shot text classification, integrating semantic representations with reinforcement learning for adaptive decision-making. Unlike traditional zero-shot approaches that rely solely on direct prediction, the proposed method incorporates uncertainty estimation $H(p)$ to guide the selection of actions, enabling the model to predict, query, or abstain depending on its confidence. By leveraging Sentence-BERT embeddings for semantic matching and a reinforcement learning policy optimized using Proximal Policy Optimization (PPO), the framework introduces a dynamic and interactive classification process. This allows the model to handle ambiguous inputs more effectively by requesting additional

information, when necessary, rather than making unreliable predictions. The proposed approach is expected to improve both classification accuracy and decision reliability, particularly in real-world scenarios where incomplete or ambiguous data is common.

ACK

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT)(RS-2025-00559546).

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